

Sunghyun Park

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“AI for Understanding the World, bridging science and technology to explore the unknown.”

Education



Yonsei University
M.S. IN ARTIFICIAL INTELLIGENCE

Seoul, South Korea
Sep. 2026 – Present (Expected Aug. 2028)



Yonsei University
B.S. IN BIOCHEMISTRY AND COMPUTER SCIENCE (DOUBLE MAJOR)

Seoul, South Korea
Mar. 2020 – Aug. 2026

- GPA : 3.94/4.3 (Cumulative), 4.03/4.3 (CS Major)
- Oct. 2022- Jul. 2024 : Compulsory Military Service, Served as a sergent in R.O.K Air Force.

Research Interest

My research interest is in the learning dynamics of foundation models, with an emphasis on understanding how optimization shapes representation learning, generalization, and forgetting. I am particularly interested in developing mathematical and empirical frameworks for analyzing the dynamics of large language models during training. My current interests center on two directions: (1) **optimization dynamics** in post-training algorithms such as preference optimization and alignment, and (2) **machine unlearning**, focusing on how information is retained, forgotten, and redistributed throughout the optimization process.

Optimization Machine Unlearning LLM

Experience

Artificial Intelligence and Information Systems Laboratory (AI-ISL) (Prof. Albert No)

GRADUATE RESEARCHER

Seoul, South Korea
July. 2026 - Present

Undergraduate Research, Mechanobiology Lab (Prof. Hyunkyu Choi)

RESEARCH INTERN

Seoul, South Korea
Dec. 2024 - Feb. 2025

- Conducted high-resolution microscopy (TIRF, RICM) analysis of biological samples and gained hands-on experience with biophysical experimental techniques.
- Applied Unsupervised Machine Learning techniques to classify cells on their orientation.

Extracurricular Activity



PoolC (Yonsei Programming Club)
MEMBER

Seoul, South Korea
Mar. 2024 - PRESENT

- Participated in AI-related paper studying groups.
- Organized seminars on the mathematical foundations of deep Learning - including information theory, optimization and Reinforcement learning.



Seoul-Learn (Mentoring Program)
MENTOR

Seoul, South Korea
Mar. 2024 - PRESENT

- Guided middle and high school students in mathematics learning and academic planning.
- Provided guidance on choosing a major-overview of CS/AI and any science major curricula, prerequisites and study habits.



Biochemistry Student Academic Club
MEMBER

Seoul, South Korea
Mar. 2020 - Mar. 2022

- Participated in collaborative study groups on core biochemistry subjects.
- Engaged in study of Organic Chemistry, focusing on reaction mechanisms through problem-solving sessions.
- Conducted group discussions on Immunology, deepening understanding of immune system functions and disease mechanisms.

Projects

Prompt Optimization with OPRO and Evolutionary algorithm

Seoul, South Korea

LLM COURSE PROJECT

Jul. 2025

- Implemented a prompt optimization loop combining **OPRO with an evolutionary algorithm** (selection, crossover, mutation).
- Used an LLM scorer to evaluate candidates and drive self-rewriting and selection.
- Evaluated on **GSM8K** with **Llama-3.1-8B-Instruct** and **Mistral-7B-Instruct**.

LLM Application Prompt Engineering

Reducing WebOS Boot Latency through Build and Boot-Time Optimization

Seoul, South Korea

YONSEI CAPSTONE DESIGN

Jun. 2025

- Analyzed **function call traces** using **fttrace** to identify critical bottlenecks in WebOS boot process.
- Distinguished kernel-level and user-level execution delays, contributing to optimization of binary layout and page fault handling.

Operating System Embedded System Boot-Time Optimization

Force-Driven Mechano-Typing of B Cells: Multi-Channel Image Analysis of CD40-CD40L Interactions Using autoencoder

Seoul, South Korea

UNDERGRADUATE RESEARCH INTERN

Dec. 2024 - Feb. 2025

- Built a dataset with multi-threshold molecular tension probe sensors and microscopy for CD40-CD40L mechanics in B cells and automated single-cell segmentation.
- Compressed each cell image data into a low dimension and estimated local intrinsic dimension for each cell data.
- Visualized the global landscape with **UMAP** and conditional-wise trajectories.

AI For Science Immunology Computer Vision

Honors & Awards

DOMESTIC

2021 Honors Award, Yonsei University

Seoul, South Korea

“Honors Award” is granted to students with outstanding academic achievement, typically ranking in the top 10%.

Relevant Coursework

CS/AI

Discrete Mathematics, Data Structures, Introduction of Computer science, Computer Architecture, Linear Algebra, **Machine Learning**, Algorithm Analysis, Computer Network, **Computer Vision**, Operating System, **Large Language Model**, Cloud Computing, Introduction of Deep Learning, **Mathematics for Deep Learning**, **Distributed Learning and inference**, **Reinforcement Learning**, **Data privacy**

Biochem

Biochemistry, Organic Chemistry, Neuroscience, **Biophysical chemistry**, Biological Statistics, Stem Cell Biology

* Courses in **bold** indicate areas of particular interest.

Skills

DevOps AWS, Docker

Programming Python, JAVA, C/C++, MATLAB, LaTeX

Framework Pytorch, OpenCV, PyMOL

Languages Korean (Native) , English (Intermediate)